WORKED SOLUTION

STEP II 2003, Question 14

Mechanics • Hypothesis testing

STEP II 2003, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q14

The key idea

The number of throws needed to get the first 6 has a geometric distribution, and its tail has a beautifully simple form: the event "more than M throws" is just "the first M throws all fail". So both error probabilities are pure Mth powers, and eliminating M between them is a matter of taking logarithms.

Solution

The two error probabilities

Model the throws as independent, each giving a 6 with the same probability p. Then X, the number of throws up to and including the first 6, satisfies

$$X > M \iff \text{none of the first } M \text{ throws is a 6}, \qquad \text{so} \qquad \mathrm{P}(X > M) = (1-p)^M,$$

using independence of the throws. (There is no need to sum the geometric series $\sum p(1-p)^{k-1}$: thinking about the event directly is quicker.)

The test accepts H_1 exactly when $X > M$. So α, the probability of accepting H_1 when in fact $p = B$, is

$$\alpha = (1-B)^M,$$

and β, the probability of accepting H_0 (that is, $X \leq M$) when in fact $p = A$, is

$$\beta = 1 - \mathrm{P}(X > M \mid p = A) = 1 - (1-A)^M.$$

Eliminating M

We want to write $(1-A)^M$ as a power of $\alpha = (1-B)^M$. Since $0 < B < 1$, $\ln(1-B)$ is defined and non-zero, so taking logarithms of α gives

$$M = \frac{\ln \alpha}{\ln(1-B)}.$$

Then

$$(1-A)^M = \mathrm{e}^{M\ln(1-A)} = \exp\left(\frac{\ln(1-A)}{\ln(1-B)}\ln\alpha\right) = \alpha^K, \qquad \text{where} \quad K = \frac{\ln(1-A)}{\ln(1-B)}.$$

Hence

$$\beta = 1-\alpha^K, \qquad K = \frac{\ln(1-A)}{\ln(1-B)},$$

and K depends only on A and B, not on M, as required.

Ask yourself What can we say about the size of K?

Since $0 < A < B < 1$ we have $1 > 1-A > 1-B > 0$, so both logarithms are negative and $\ln(1-B) < \ln(1-A) < 0$. Dividing, K is positive and

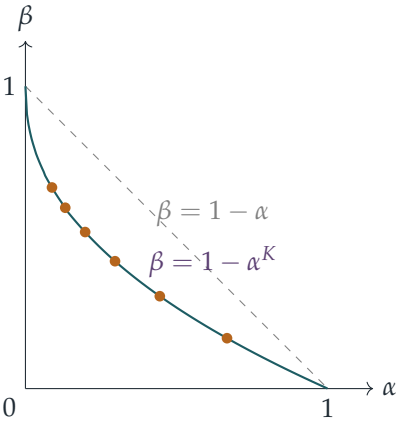
$$0 < K < 1.$$

This is what fixes the shape of the graph.

The graph of β against α

As M runs through $1, 2, 3, \ldots$, $\alpha = (1-B)^M$ takes values in $(0,1)$, decreasing towards 0; strictly the graph is a sequence of points, all lying on the curve $\beta = 1-\alpha^K$, $0 \le \alpha \le 1$. For that curve:

- it passes through $(0,1)$ and $(1,0)$;
- $\dfrac{\mathrm{d}\beta}{\mathrm{d}\alpha} = -K\alpha^{K-1} < 0$, so β decreases as α increases: making one error less likely makes the other more likely;
- since $K - 1 < 0$, the gradient tends to $-\infty$ as $\alpha \to 0^+$ (the curve leaves $(0,1)$ vertically) and equals $-K$, which is between -1 and 0, at $\alpha = 1$;
- $\dfrac{\mathrm{d}^2\beta}{\mathrm{d}\alpha^2} = K(1-K)\alpha^{K-2} > 0$, so the curve is convex;
- for $0 < \alpha < 1$ and $0 < K < 1$ we have $\alpha^K > \alpha$, so $\beta < 1-\alpha$: the curve lies below the straight line joining $(0,1)$ to $(1,0)$.



The curve is drawn for $A = \frac{1}{6}$, $B = \frac{1}{3}$ (so $K \approx 0.45$), with the points for $M = 1, \ldots, 6$ marked; increasing M moves the point up and to the left.

Checks

- Take $A = \frac{1}{6}$, $B = \frac{1}{3}$, $M = 4$. Directly, $\alpha = (\frac{2}{3})^4 \approx 0.1975$ and $\beta = 1 - (\frac{5}{6})^4 \approx 0.5177$. With $K = \ln \frac{5}{6} / \ln \frac{2}{3} \approx 0.4497$, $1 - \alpha^K \approx 0.5177$ too. A simulation of 2×10^5 runs of the test gave 0.198 and 0.518.
- Extreme cases behave sensibly: if A is close to B then $K \approx 1$ and $\beta \approx 1 - \alpha$, so the test is little better than ignoring the throws and deciding at random; if A is tiny, $K \approx 0$ and β can be made small for any reasonable α.

Common slips

- Mixing up which hypothesis is which: here H_0 is the *larger* value B, so a long wait for a 6 is evidence against H_0.
- Getting K upside down, $\ln(1 - B) / \ln(1 - A)$, which is greater than 1; the check $\beta < 1 - \alpha$ (any sensible test beats guessing) catches this.

Beyond STEP

The curve of β against α is the trade-off curve between the two types of error, closely related to the ROC curve used throughout statistics and machine learning. The Neyman–Pearson lemma shows that, for two simple hypotheses like these, a test based on the likelihood ratio gives the smallest possible β for each α; here the likelihood ratio is monotonic in X, so the test "reject H_0 when X is large" is in fact optimal. For more on these ideas, see Grimmett and Welsh, *Probability: An Introduction*, as a foundation, then any first course in mathematical statistics.

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